Fast Eyes, Sharp Eyes, or Good Eyes? What Links Visual Perception to Higher-Order Cognition

Shanglin Yang[1], Jeffrey N. Rouder[1], & Rean Du[1]

[1] University of California, Irvine

Author Note

Version 1, September, 2026.

Correspondence concerning this article should be addressed to Jeffrey N. Rouder. E-mail: jrouder@uci.edu

Fast Eyes, Sharp Eyes, or Good Eyes? What Links Visual Perception to Higher-Order Cognition

Introduction

One of the most striking findings in individual-differences research is that performance on simple perceptual tasks predicts performance on higher-order cognitive tasks such as working memory and fluid intelligence tasks (Jastrzębski, Kroczek, & Chuderski, 2021; Robison, Campbell, Garner, Sibley, & Coyne, 2026; Tsukahara, Harrison, Draheim, Martin, & Engle, 2020). The finding is old, perhaps first explored by Galton (1883) and Spearman (1904). Perhaps the best documented link between perceptual ability and higher-order cognition comes from the inspection time (IT) task. The classic version of the task is shown in Figure 1A. Here, individuals briefly view a simple figure and make a perceptual judgment. The brief display of the simple figure is followed by a pattern mask such that the visual information is available for a limited time. Consequently, accuracy is not at ceiling (Vickers, Nettelbeck, & Willson, 1972). Individuals with higher accuracy on these inspection-time tasks tend to score higher on tests of intelligence (Deary & Stough, 1996; Nettelbeck & Lally, 1976), with a meta-analytic correlation of about .3 across 92 studies (Grudnik & Kranzler, 2001; Kranzler & Jensen, 1989).

Why should accurate judgement about brief displays be related to high performance in working memory and IQ tasks? The standard answer is based on *processing speed*. People who do well on the IT task are good at extracting visual information quickly. They have fast visual processing speed. The claim then is that processing speed is not domain specific but is a general phenomenon that affects many cognitive processes (Jensen, 2006; Salthouse, 1996; Vernon, 1983). Individuals with fast processing speed do well in working memory and IQ tasks because it is general. The processing speed account of the correlation makes sense. Yet, we worry here about the claim that inspection time indexes processing speed per se. Certainly, it is plausible that individual differences in inspection time tasks

reflect processing speed, but, on the other hand, they may reflect factors that do not have a temporal basis. We refer to these as spatial factors, in contrast to the temporal factor of processing speed: the ability to resolve the target clearly and to filter it from the mask. In the modern literature, perceptual abilities that are not time limited also correlate with intelligence and working memory (Deary, Bell, Bell, Campbell, & Fazal, 2004; Mikellidou, Lambrou, Georgiou, & Avraamides, 2023; Mosing, Pedersen, Madison, & Ullén, 2014; Tsukahara et al., 2020). A good example of this correlation comes from Tsukahara et al. (2020) who measured how well individuals could discriminate between the loudness of tones, the pitch of tones, the length of lines, and the area of circles. These discrimination abilities correlated between 0.16 and 0.53 with progressive matrix reasoning tasks. These findings do not rule out a processing speed explanation *per se* as these tasks may require speediness in maintaining representations across different fixations and attentional moments for comparison.

In practice, it seems difficult to determine if the correlation between perception and higher-order cognition is mediated by temporal or spatial factors. The reasons for these difficulties may be seen in the recent study by Jastrzębski et al. (2021) who administered a series of IT tasks, a series of sensory discrimination tasks, and a battery of working memory and intelligence tasks to the same participants. There are both statistical and methodological reasons for the difficulty. The statistical reason is that the correlations form an almost perfect triangle of values: the latent inspection-time factor correlated about .63 with the latent sensory-discrimination factor, and each correlated substantially with working memory ($r = .68$ between inspection-time and working memory, and $r = .66$ between sensory-discrimination and working memory). Yet such a triangle tells us little about why perception relates to cognition. The methodological problem is that the stimuli in inspection time and sensory discrimination are not matched or yoked. In Jastrzębski et al. (2021), they used partly different judgments (e.g., length vs. brightness) and partly different stimuli (e.g., lines vs. rectangles) for two kinds of tasks, so their relation to

cognition may reflect specific judgments or stimuli rather than temporal or spatial factors.

Our goal is to put spatial and temporal factors in contrast and ask which factor better mediates the correlation with higher-order cognition. For the purposes herein, we distinguish between *fast eyes* and *sharp eyes*. Individuals with fast eyes process visual information quickly; people with sharp eyes process spatial information well in the presence of visual noise. If the fast-eyes account is correct, the relation between IT and cognitive performance arises because the temporal mask limits the time available to process the target. Remove this time limit, and the relation should disappear. On the other hand, if the sharp-eyes account is correct, then the relation reflects the ability to see under impoverished visual conditions. Improve the visual conditions, and the relations should disappear.

These diverging predictions may be tested in a highly controlled environment where only the type of masking differs. In the *temporal-masking* versions of the tasks, the stimuli were relatively clear and the key manipulation was the time between stimulus and a following patterned temporal mask. In the *spatial-masking* versions, the stimulus was presented in stimulus-tuned noise, and the stimulus was displayed for a sufficient time. In the temporal masking version, the duration of information is manipulated and individual differences rely more on processing speed (fast eyes) than acuity (sharp eyes). In the spatial masking version, the amount of noise is manipulated and individual differences rely more on acuity than processing speed. Key in this setup is that all other aspects of the stimulus are fairly constant across temporal-masking and spatial-masking versions. Thus, the experiment has the potential to discriminate between the fast-eyes and sharp-eyes hypotheses.

We constructed six different visual perception tasks by crossing three stimulus types with the two mask types. The two mask types are whether the stimulus is degraded by temporal masking or by spatial masking. The three stimulus types are whether the target

is a "pi" stimulus which is standard in inspection time tasks, a letter stimulus, or a gabor grating. The pi stimulus consists of two vertical segments connected by an upper horizontal segment, and the individual judges whether the left or right leg is longer. Letter stimuli are just these, and the individual judges which letter is presented. Gabor stimuli have an orientation that is either -45 degrees from the vertical (left tilting) or +45 degrees from the vertical (right tilting). Individuals judge whether the gabor orientation is left or right tilting.

The expected data patterns are as follows: if the fast-eyes hypothesis holds, then the three temporal-masking tasks should cluster well and the combined latent variable should correlate well with higher-order tasks. Conversely, if the sharp-eyes hypothesis holds, then the three spatial-masking tasks should cluster well and the combined latent variable should correlate well with higher-order tasks. Yet, there are data patterns that do not accord with either hypothesis. Perhaps, there is no special concept of fast or sharp eyes. An alternative is that some people have better vision than others, and that those with better vision tend to do better on the higher-order cognitive functions. We call this the *good-eyes* hypothesis.

Given the large literature on the correlation between inspection time and IQ, we ran this experiment to confirm that processing speed was crucial. We fully expected the temporal-masking tasks to cluster and the shared variance among them to drive the correlation between perception and higher-order cognition. We were quite wrong in these expectations. To foreshadow, we find that the distinction between spatial and temporal masking accounts for very little of the variance across individuals. A factor that accounts for more variance is stimulus type. It seems that people vary more across gabors, letters, and pi stimuli than they do across spatial and temporal masks. Moreover, it seems that it is what is shared across the stimulus types that drives the correlation with higher-order cognition. And this fact provides the best evidence that neither spatial nor temporal masking is privileged in relating to higher-order cognitive functions. It seems the good-eyes hypothesis, our add on hypothesis for completeness, best describes the data.

Experimental Method

The experimental battery consisted of 10 tasks that was run on 83 individuals. Six of the ten tasks are discussed above and serve as the perceptual part of the battery. These six are formed by crossing the type of stimulus (pi, letter, gabor) with type of masking (spatial, temporal). Of the remaining four tasks, three are span tasks to assess working memory and a fourth is a matrix reasoning task to assess fluid intelligence.

Participants

Eighty-three undergraduate students at the University of California, Irvine, participated in exchange for $50 in total compensation. Four individuals withdrew before completing all sessions, leaving a final sample of 79. All individuals reported normal or corrected-to-normal vision and provided informed consent. The study was approved by the Institutional Review Board of the University of California, Irvine (Study ID: STUDY00000064).

Apparatus

Participants were tested individually in a dimly lit cubicle, seated approximately 100 cm from a 24-inch BenQ LED gaming monitor (1920 × 1080 pixels, 165 Hz refresh rate, 6.06 ms per frame). Stimuli were generated with PsychoPy3 (Peirce et al., 2019) in Python 3.8, running on a PC with Lubuntu Linux 22.04.

Tasks

Task 1: Pi Stimuli with temporal Masking. Participants completed a two-alternative forced-choice length discrimination task on a Π-shaped (Pi) target, as shown in Figure 1A. Each trial began with the presentation of a fixation cross, which was displayed for 485 ms. It was followed by a centrally presented Π stimulus in which one of

the two vertical "legs" was longer than the other. After a variable stimulus onset asynchrony (SOA), the target was replaced by a dynamic sequence of five masks, each presented for 48.5 ms. Each mask was composed of random oblique line segments forming a Π-like clutter. Short line elements (randomized in length and orientation) were placed in two vertical columns aligned with the legs of the Π and around the crossbar region. Participants judged whether the left or right leg was longer by pressing designated keys on the keyboard. The SOA was manipulated adaptively on a trial-by-trial basis for each participant.

Task 2: Letter Stimuli with temporal Masking. Participants completed a 9-choice letter identification task as shown in Figure 1B. The presented letters were "A", "S", "D", "F", "G", "H", "J", "K", and "L", that is, those that are on the second row of a standard American English QWERTY keyboard. Each trial began with the presentation of a fixation cross, which was displayed for 485 ms. It was followed by a centrally presented target letter. After a variable SOA, the target was replaced by a dynamic sequence of five masks, each presented for 48.5 ms. The masks were FFT-filtered noise with spatial frequencies matching the collection of letters (Parish & Sperling, 1991). Participants identified the letter by pressing the corresponding key on the keyboard. The SOA was manipulated adaptively on a trial-by-trial basis for each participant.

Task 3: Gabor Stimuli with temporal Masking. Participants completed a two-alternative forced-choice orientation discrimination task as shown in Figure 1C. Each trial began with the presentation of a fixation cross, which was displayed for 485 ms. It was followed by a centrally presented Gabor patch with a random phase and an orientation of $+45°$ or $-45°$ relative to vertical. The patch subtended approximately $8.1° \times 8.1°$ of visual angle and had a spatial frequency of approximately 1.2 cycles per degree, with a Gaussian envelope of standard deviation of approximately 1.0°. After a variable SOA, the target was replaced by a dynamic sequence of five white noise masks, each presented for 48.5 ms. Participants judged whether the grating was tilted leftward ($-45°$) or rightward ($+45°$) by

pressing designated keys on the keyboard. The SOA was manipulated adaptively on a trial-by-trial basis for each participant.

Task 4: Pi Stimuli with Spatial Masking. This task matched the temporal Pi-leg discrimination task (Task 1), as shown in Figure 2A, except that the mask was presented together with the target rather than after it. Each trial began with the presentation of a fixation cross, which was displayed for 485 ms. It was followed by the Π target superimposed on a Π-like clutter mask constructed in the same way as in Task 1, and this target-mask combination was presented for 485 ms. Participants judged whether the left or right leg was longer by pressing designated keys on the keyboard. Instead of the SOA, the leg-length difference was manipulated adaptively on a trial-by-trial basis for each participant.

Task 5: Letter Stimuli with Spatial Masking. This task matched the temporal letter identification task (Task 2), as shown in Figure 2B, with the same nine letters and response keys, except that the target letter was embedded in noise rather than followed by a mask. Each trial began with the presentation of a fixation cross, which was displayed for 485 ms. It was followed by a target letter presented in FFT-filtered noise with spatial frequencies matching the collection of letters, and this stimulus-noise combination was presented for 485 ms. Instead of the SOA, the contrast of letter-to-noise was manipulated adaptively on a trial-by-trial basis for each participant.

Task 6: Gabor Stimuli with Spatial Masking. This task matched the temporal Gabor orientation discrimination task (Task 3), as shown in Figure 2C, except that the Gabor patch was embedded in noise rather than followed by a mask. Each trial began with the presentation of a fixation cross, which was displayed for 485 ms. It was followed by a Gabor patch with the same size, spatial frequency, and envelope as in Task 3, with a random phase and an orientation of $+45°$ or $-45°$ relative to vertical, presented in white noise. This stimulus-noise combination was presented for 485 ms. Participants judged

whether the grating was tilted leftward (−45°) or rightward (+45°) by pressing designated keys on the keyboard. Instead of the SOA, the Gabor contrast was manipulated adaptively on a trial-by-trial basis for each participant.

Task 7: OSPAN. Participants completed an automated OSPAN task (Unsworth, Heitz, Schrock, & Engle, 2005) combining simple math operations with working memory recall for letter sequence. Each set contained 3–7 letters, with each set size presented three times, yielding 15 memory sets in total. Before the main task, individuals completed three practice sets of size three to familiarize themselves with the procedure. In the main task, set sizes 3–7 were each presented three times in randomized order (15 sets total). Within each set, trials alternated between (i) a math operation screen (e.g., "6 + 2 = ?"), after which the individual clicked to continue, and (ii) a proposed answer screen showing a single digit; individuals judged whether the digit correctly solved the operation by clicking true/false. A single uppercase letter was then shown for 800 ms, and the sequence continued until the set ended. At recall, individuals clicked the letters in serial order from a matrix. For each set, individuals earned points equal to the set size only if (a) all math operations were solved within the individual's mean solving time in the practice trials plus 2.5 SD, (b) no math errors occurred for that set, and (c) all letters were recalled in the correct order. Sets failing any criterion scored 0. The final OSPAN score was the sum of points across all sets.

Task 8: CSPAN. Participants completed a CSPAN task (Conway et al., 2005) combining serial counting with working memory recall for number sequence. Each display contained a set of target items (green circles) intermixed with distractors (yellow circles). individuals counted the number of targets in each display, after which the individual clicked to continue to the next display. At recall, individuals were prompted to recall the sequence of counts (e.g., "3–5–2") in the exact order they were encountered. Responses were made by clicking the digits from a matrix. The procedure began with three practice sets of size 2. The main task then presented set sizes 3–7, each three times in randomized

order (15 sets total). A set contributed points equal to its size only if the final recall sequence was accurate in both values and order. The final CSPAN score was the sum of points across all sets.

Task 9: SSPAN. Participants completed a SSPAN task adapted from the Corsi block-tapping paradigm to assess both storage and processing aspects of spatial working memory (Pahor, Seitz, & Jaeggi, 2022). On each trial, cartoon gophers appeared sequentially in one of 12 possible screen locations, each presented for 1,500 ms with a 500 ms inter-stimulus interval. individuals were required to reproduce the sequence by tapping the corresponding locations in the exact order. Between each gopher presentation individuals performed a secondary sorting task, dragging an item to the left or right, thereby imposing concurrent processing demands and limiting reliance on mnemonic strategies. Sequence length ranged adaptively from 2 to 10 items, increasing or decreasing in difficulty depending on accuracy. The final SSPAN score was the total successfully recalled items across all sets.

Task 10: UCMRT. Participants completed the UCMRT (Pahor, Stavropoulos, Jaeggi, & Seitz, 2019), a measure of fluid intelligence modeled after Raven's Progressive Matrices. Each trial presented a 3×3 matrix of abstract visual patterns with the bottom-right cell missing, and individuals selected the option that best completed the matrix from eight alternatives displayed below. UCMRT was chosen because it was especially developed for young adults with above-average performance on matrix reasoning tasks, providing a more sensitive and efficient assessment of individual differences in fluid intelligence in this population. The task consisted of 23 problems of varying difficulty, including 2 two-relation items, 3 three-relation items with a single transformation, 6 three-relation items with two transformations, 6 three-relation items with three transformations, and 6 logic items. Performance was scored as accuracy, defined by the number of correctly solved matrices.

Procedure

The experiment consisted of three sessions of approximately one hour each, conducted on separate days within one week. Each session consisted of three phases, denoted here Phase I, Phase II, and Phase III. Phase I consisted of the six perceptual tasks (Task 1-6). Each task was shown in a single block of 50 trials. All six tasks were performed for 50 trials each, and the ordering of the tasks was random across days and participants. Phase III was the same; and across Phase I and Phase III, 100 trials per perceptual task was performed. These phases were repeated on all three sessions, yielding 300 trials per participant per perceptual task. This number is quite large compared to other individual difference studies using perceptual tasks. In these phases and blocks, the difficulty was manipulated in an adaptive fashion. Two staircases were run interleaved; one was a 2D-1U staircase with an asymptotic limit of .707; the other was a 3D-1U staircase with an asymptotic limit of .794 (Levitt, 1971). The first masking block served as practice and was excluded from analysis, leaving 250 trials per task (125 per staircase) and 1,500 masking trials per individual for analysis.

Phase II of each session was reserved for the working memory and matrix reasoning tasks. The OSPAN task was performed in Phase II on Session 1; the CSPAN task was performed likewise on Session 2; the SSPAN and UCMRT were performed on Session 3.

Summary-Score Analysis and Results

We analyzed the data in two ways. The first is a conventional analysis of summary scores. For each individual and task, we computed a single score, screened these scores for outliers, and examined the correlations among tasks. This analysis is simple and transparent, and it lets readers see the data directly. The second is a hierarchical psychometric model fit to the trial-level data. It estimates the relations among tasks while accounting for trial-to-trial noise, so that the estimated correlations are not attenuated and

their uncertainty reflects both the sampling of trials and the sampling of individuals. The model also decomposes individual differences into general, masking-specific, and stimulus-specific sources, each of which is related to cognitive performance. The summary-score analysis serves as a descriptive first look, and the hierarchical model provides a more complete accounting of the data.

The first step in the summary-score analysis is constructing a single score that summarizes performance on each task for each individual. For the three temporal masking tasks, the main independent variable is the stimulus onset asynchrony between the target and the mask, which was manipulated adaptively in staircase fashion. For the three spatial masking tasks, the main independent variables are the leg-length difference of the Pi stimulus, the target weight in the letter–noise blend, and the contrast of the Gabor patch, each of which was likewise manipulated adaptively. To construct these summary scores, the mean duration across the 250 analyzed trials was computed. For the four cognitive tasks, Tasks 7-10, the scoring method is standard to the task and described above.

The main target of the summary-score analysis is the correlations between tasks across individuals. The main concern in data quality is that some scores exert unreasonably high leverage on these correlations. Figure 3 shows two scatterplots of individual scores, each plotting one task against another. Both include the same outlying individual. In Figure 3A, this individual does not change the correlation appreciably; in Figure 3B, the individual does. We decided the best approach was to discard any score on a task that was more than 4 standard deviations from the task mean, regardless of its effect on the correlations. In total, data from 2 task-by-individual combinations (out of a total of 790 combinations) were discarded. Readers can confirm for themselves that the 4 SD criterion is appropriate; they can also explore the effects of other criteria, which, while affecting the values of specific correlations, do not affect the main conclusions herein.

The surface-level results are shown in Figure 4. Panel A shows the correlations

between tasks, and these are about .29 on average: .32 among the six masking tasks, .39 among the four cognitive tasks, and .25 between masking and cognitive tasks. To interpret the pattern of these surface correlations vis-a-vis the hypotheses, we simply replot the correlations between the temporal-masking tasks and the higher-order cognitive tasks and the correlations between the spatial-masking tasks and the higher-order cognitive tasks. The two submatrices are shown in Panel B, and to our surprise the correlations are larger for spatial-masking tasks than for temporal-masking tasks (mean $r = .30$ vs. .19). Thus, at the level of surface correlations, neither type of masking is clearly privileged in its relation to higher-order cognition.

To see directly how the tasks group together, we used hierarchical clustering with average linkage (Sokal & Michener, 1958), as implemented in the hclust function in R, with $1 - r$ as the distance between two tasks. The results are shown in Figure 4C. In this representation, tasks that are more similar are presented together lower in the hierarchy. In all cases, the lower level of the most similar perceptual tasks are arrange by stimulus type—gabor grating, letters, and pi stimuli—and not by mask type. For example, temporals masking in the Gabor stimuli are quite disimilar than temporals masking in the pi stimuli. Perhaps the only unexpected finding then is that the pi stimuli is disimilar to the other perceptual tasks even though the only difference is the type of stimulus. As it turns out, when a more complete accounting of the data through hierarhical modeling is performed next, this unexpected finding does not hold.

Hierarchical psychometric model analysis

The summary-score analysis has three limitations. First, the correlations are attenuated, because each score contains trial-to-trial noise in addition to an individual's true ability (Rouder & Haaf, 2019; Spearman, 1904). Second, their uncertainty is miscalculated, because it reflects only the sampling of individuals and not the sampling of trials. Third, the correlations alone cannot distinguish among the fast-eyes, sharp-eyes, and

good-eyes hypotheses, because these hypotheses concern which source of shared variance carries the relation to cognitive performance. To address all three, we estimated individual thresholds and their relations simultaneously from trial-level data with a hierarchical psychometric model (Rouder & Mehrvarz, 2026). Because the model accounts for trial-to-trial noise, the estimated correlations are not attenuated and their uncertainty reflects both sources of variability; and because the latent level decomposes individual differences into general, masking, and stimulus factors, the model directly tests which source relates to cognitive performance.

The hierarchical model has three levels. The top level is termed here the *data level,* and it describes how trial data in the perceptual tasks reflect individuals' true perceptual abilities. The next level, termed the *measurement level*, and it describes how these individual true scores reflect a factor structure where the factors are type of stimulus, type of mask, general visual ability, and higher-order cognitive ability. The final level, termed the *structural level*, and it describes the association between factors, and in particular, it describes how the cognitive abilities are related to the perceptual abilities. We describe each level in turn.

Data level. To specify levels, we need introduce some notation. The data level is for the perception tasks and it describes how trial-level performance reflects true perceptual abilities. Let $i = 1, \ldots, I$ denote an individual, let $j \in \{1, 2, 3, 4, 5, 6\}$ denote the perception task, and let $k = 1, \ldots, K_{ij}$ denote trial for the ith individual on the jth task. Let y_{ijk} denote the corresponding accuracy, $y_{ijk} = 0, 1$ for error or correct responses, respectively. A lognormal psychophysical data model (Rouder & Mehrvarz, 2026) is

$$y_{ijk} \sim \text{Bernoulli}(\Phi(\beta_j\,(s_{ijk} + \theta_{ij}))\,,$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. The variable s_{ijk} denotes the **strength** of the strimulus defined as follows: On each trial, there is a natural

stimulus intensity variable. For example in temporal masking tasks, this intensity is stimulus duration measured in milliseconds. The strength is the log of this duration. Taking the log has two degrees of freedom, the base and a multiplier which serves as a referent. To set these degrees of freedom in a consistent way, we simply took the z-score of log intensity where setting mean and variance of the log scores is equivalent to picking a base and referent. Hence, strength here is standardized logarithm of a natural intensity variable. Please see Rouder and Mehrvarz (2026) for additional discussion.

There are two parameters in the above pshophysical model: θ_{ij} and β_j. The parameter θ_{ij} denotes the latent ability of individual i on task j, expressed on the same scale as s_{ijk}, and β_j denotes the rate of psychometric function on strengths for task j.

Measurement level. The measurement level describes how true abilities reflect specific factors. Each individual's perceptual abilities are characterized by the vector $\boldsymbol{\theta}_i = (\theta_{i1}, \ldots, \theta_{i6})'$. Individual's also performed in the cognitive tasks but the scoring rules are standard and do not reflect a trial structure. Let $\boldsymbol{c}_i = (c_{i1}, \ldots, c_{i4})'$ be the vector of each individuals' cognitive abilities. A complete vector of abilities on the ten tasks, denoted $\boldsymbol{\gamma}_i$ are:

$$\boldsymbol{\gamma}_i = (\boldsymbol{\theta}_i', \boldsymbol{c}_i')'.$$

The next task is define a reduced set of factors that capture the design and hypotheses. We specify seven factors as follows: a general visual factor, a factor that reflects temporal masking, a factor that reflects spatial masking, a factor that reflects Pi stimuli, a factor that reflects Letter stimuli, a factor that reflects Gabor stimuli, and a factor that reflects higher-order cognition.

The standard factor model is used to relate factors to abilities:

$$\boldsymbol{\gamma}_i = \boldsymbol{\mu} + \boldsymbol{\Lambda}\,\boldsymbol{\eta}_i + \boldsymbol{\epsilon}_i, \qquad \boldsymbol{\epsilon}_i \sim \mathcal{N}_{10}(\mathbf{0}, \boldsymbol{\Delta}), \qquad \boldsymbol{\Delta} = \mathrm{diag}\left(\delta_1^2, \ldots, \delta_{10}^2\right),$$

where $\boldsymbol{\mu}$ captures the mean of each task, $\boldsymbol{\Lambda}$ is a 10×7 matrix of factor loadings, $\boldsymbol{\eta}_i$ is a 7-dimensional vector of latent factors for the ith individual, $\boldsymbol{\epsilon}_i$ is a residual with task-dependent variance. The key constraint comes from constraints on the factor loadings, which is given as follows:

$$\boldsymbol{\Lambda} = \begin{array}{c} \begin{array}{ccccccc} \text{Visual} & \text{Temporal} & \text{Spatial} & \text{Pi} & \text{Letter} & \text{Gabor} & \text{Cognitive} \end{array} \\ \begin{pmatrix} \lambda_{11} & \lambda_{12} & 0 & \lambda_{14} & 0 & 0 & 0 \\ \lambda_{21} & \lambda_{22} & 0 & 0 & \lambda_{25} & 0 & 0 \\ \lambda_{31} & \lambda_{32} & 0 & 0 & 0 & \lambda_{36} & 0 \\ \lambda_{41} & 0 & \lambda_{43} & \lambda_{44} & 0 & 0 & 0 \\ \lambda_{51} & 0 & \lambda_{53} & 0 & \lambda_{55} & 0 & 0 \\ \lambda_{61} & 0 & \lambda_{63} & 0 & 0 & \lambda_{66} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{77} \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{87} \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{97} \\ 0 & 0 & 0 & 0 & 0 & 0 & \lambda_{10,7} \end{pmatrix} \end{array}.$$

Here are how the factor loadings work. Each row is for a particular task, and the first row is for Task 1, the Pi stimulus with temporal masking. The first factor is for perceptual tasks and there is a positive valued loading, λ_{11} for this task and factor. The second factor and third factors are for temporal and spatial masking tasks, respectively, and whereas this task is temporal and not spatial, there is a positive loading λ_{12} and a zero loading for λ_{13}. The fourth, fifth and sixth factors are for pi, letter, and gabor stimuli, respectively, and consequently, there is a positive loading λ_{14} and zero loadings for λ_{15} and λ_{16}. The last factor is for higher-order cognitive tasks, and hence, there is a zero loading for λ_{17}. The remainder of the rows are valuated analogously.

The above factor loadings define a standard bifactor model for the six perceptual

tasks. The first factor is general visual ability across the six tasks; the next five factors define specific effects of masks and stimuli. The next level leverages this bifactor representation to provide specific assessments of the fast-eye, sharp-eye, and good-eye hypotheses as discussed next.

Structural level. The structural model describes how cognitive abilities relates to the perceptual abilities across individuals. The key constraint is on $\boldsymbol{\eta}_i$, each individual's 7-dimensional profile of abilities. In structural equation models, these abilities are related through regression equations. We follow a standard notation in structural equation modeling (Bollen, 1989). Readers in cognition should note that this notation, while standard, is a bit strange. The structural part of the model is given by

$$\boldsymbol{\eta}_i = \mathbf{B}\,\boldsymbol{\eta}_i + \boldsymbol{\zeta}_i, \qquad \boldsymbol{\zeta}_i \sim \mathcal{N}_7(\mathbf{0}, \mathbf{I})$$

The strange-looking part is $\boldsymbol{\eta}_i = \mathbf{B}\,\boldsymbol{\eta}_i + \boldsymbol{\zeta}_i$, which at first glance seems self referential or recursive. It is not because there are strict constraints on $\mathbf{B}$, which is also known as the path matrix in SEM analysis. For our hypotheses, the appropriate specification of $\mathbf{B}$ is:

$$\mathbf{B} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ b_1 & b_2 & b_3 & b_4 & b_5 & b_6 & 0 \end{pmatrix}.$$

Only the last row contains free parameters, so the cognitive factor is the only factor that receives paths from the other factors. This specification then implies $\eta_{i7} = \sum_{\ell=1}^{6} b_\ell\,\eta_{i\ell} + \zeta_{i7}$, where ℓ indexes the six non-cognitive factors. This specification insures that the perceptual

factors are orthogonal (any correlation is accounted for by the general visual factor) and that the effects of each perceptual factor on the cognitive factor is separately indexed by $b_1, \ldots, b_6$.

How does this structural equation component capture the fast eyes, sharp eyes, and good eyes hypothesis? If cognitive abilities covary uniquely with temporal masking abilities, that is, the fast-eyes hypothesis holds, then regression weight b_2 should be large. Likewise, if cognitive abilities uniquely covary with spatial masking abilities, that is, the sharp-eyes hypothesis holds, then regression weight b_3 should be large. Finally, if masking type has little differential effect, that is the good-eyes hypothesis holds, then b_1 should be large. The regression weights b_4, b_5, and b_6 refer to the effect of stimulus type, and people who do differentially well on specific stimulus types may do well on higher-order cognition tasks.

Modelling Results

The model was estimated with Markov chain Monte Carlo sampling; software, priors, and convergence diagnostics are reported in Appendix A. All quantities reported below were computed for each posterior draw and are summarized by their posterior means and 95% credible intervals.

Manifest Correlation

The first analysis presents the model-based version of the summary-score analysis. Here, we derive the posterior manifest correlation matrix across the 10 tasks. The covariance form of this matrix, denoted $\mathbf{\Sigma}$ is a 10×10 matrix given by $\mathbf{\Sigma} = \mathbf{\Lambda Q Q' \Lambda'} + \mathbf{\Delta}$, where $\mathbf{Q} = (\mathbf{I} - \mathbf{B})^{-1}$.. The resulting values are shown in Figure 5A. There are two main differences between these model-derived values and the summary score correlations. First, the model-derived values are larger reflecting a disattenuation of trial noise. Second, the model-derived values vary more smoothly reflecting regularization from a reduced number

of factors. Figure 5B shows the submatrices for the fast and sharp eye hypotheses, and as can be seen, the cognitive factors seem somewhat better associated with spatial than temporal masks. Figure C shows the hierarchical clustering of the model-estimated correlations. The first three merges each joined the temporal and spatial versions of the same stimulus, first Pi, then Letter, then Gabor, and only after these pairs had formed did they combine with one another. In contrast, neither the three temporal masking tasks nor the three spatial masking tasks formed a cluster of their own. Instead, all six masking tasks formed one cluster, and the four cognitive tasks formed a separate cluster that joined the masking tasks only at the final step. Thus, individual differences in masking performance were organized primarily by stimulus type rather than by mask type.

Bifactor Analysis of the Perception Tasks

The six perception tasks were modeled as bifactor model with a common visual factor and unique factors for the two mask type and the three stimulus type. For each task, we may ask how much variance is accounted for by the general factor, the particular masking factor, and the particular stimulus factor. Table 1 provides the proportions. For example, the first row shows the proportion for Task 1, the temporal masking of pi stimuli. The visual factor accounts for 30% of the variance; the relevant masking factor accounts for only 4% of the variance, the relevant stimulus factor accounts for 45% of the variance; and there 21% unique to this task. Perhaps the most informative entry is the Average row, which shows that mask type accounts for very little variance in these tasks. It seems that people vary on how good they are on all tasks, and how good they are differentially on the stimulus types. This finding is in accordance with the clustering results in Figure 5C, but the degree of uncertainty for both finite numbers of individuals and trials is included.

Association between perceptual factors and higher-order cognition: A test of the fast-eyes, sharp-eyes, and good-eyes hypotheses.

The hierarchical model is designed to assess the critical hypotheses directly. The pivotal quantities are the regression weights $b_1, \ldots, b_6$ in the structural model. One highly-interpretable way of assessing these weights is to report the implied correlation between each of the six perceptual factors and the cognitive score. This correlation is given by

$$r_\ell = \frac{b_\ell}{\sqrt{\sum_{\ell'=1}^{6} b_{\ell'}^2 + 1}},$$

where r_ℓ is the correlation between the ℓth perceptual factor and the cognitive factor. The fast-eyes, sharp-eyes, and good-eyes hypotheses thus correspond to large values of r_2, r_3, and r_1, respectively. These correlation values, along with credible intervals, are shown in Table 2. The derivation of the correlation is provided in Appendix B.

To perform inference on these values, we used a Bayes factor model comparison approach. The main question is whether the perceptual factor is or is not related to the cognitive factor, and we did a separate model comparison test for each factor. The Bayes factor is a ratio—it is the probability of the data under a model with possible correlation between the specified factor and the cognitive factor relative to the probability of the data under a model with no correlation under between the specified factor and the cognitive factor. For example, for the general visual factor, we tested whether the associated correlation was 0 given free correlations for all other factors, and so on. Our computational approach was to use the Savage-Dickey ratio (Dickey & Lientz, 1970) and evaluate it from MCMC outputs using the CMDE method of (Morey, Rouder, Pratte, & Speckman, 2011). Values less than 1 indicate evidence for a null correlation; values at 1 are equivocal; values greater than 1 are evidence for a substantial correlation. Table 2 shows these Bayes factors. One factor stands out—there is a substantial correlation between the visual factor, the one that captures what is in common in all the perceptual tasks, and higher-order cognition.

The remaining factors have Bayes factors sufficiently close to one to be equivacable; the uncertainty in the data is just too great to make more certain inferences. The good-eyes hypothesis is best supported by the data.

Discussion

We asked whether the link between simple visual perceptual tasks performance and higher-order cognitive tasks performance reflects fast eyes, sharp eyes, or good eyes. By fast eyes, we mean the ability to take in visual information quickly. By sharp eyes, we mean the ability to resolve a target clearly in the presence of visual noise. By good eyes, we mean a general visual ability that is not specific to either temporal or spatial factors. To answer this question, we crossed three stimulus types with two masking procedures, so that the temporal and spatial versions of each task used the same stimulus, the same judgment, and differed only in mask type. The fast-eyes hypothesis predicted that the temporal masking tasks would cluster together and that their shared variance would carry the relation to cognitive performance. The sharp-eyes hypothesis predicted the same for the spatial masking tasks. To capture these hypotheses, we developed a hierarchical psychometric-SEM model that allow for understanding the effect of stimulus and mask type as well as decomposing the relationship between the perceptual and cognitive tasks. The substantive implication of the results are as follows:

1. Neither the fast-eyes nor the sharp-eyes hypothesis hold. Masking type accounts for little variation among the six perceptual tasks, and, moreover, does not correlate with higher-order cognitive tasks.

2. Stimulus type accounted for a sizable share of the variance. Hence, there are categories of individuals, some are good at seeing Pi stimulus, some are good at seeing letter stimulus, and some are good at seeing gabor stimulus. Even so, this factor does not correlate with higher-order cognitive tasks.

3. We included a general visual factor within a bifactor model of perceptual tasks. This factor accounts for commonalities among the six perceptual tasks. Although this factor does not account for all the variance—that is, there are stimulus-specific factors—it is the only factor that is associated with higher-order cognition. This result confirms the good-eyes hypothesis. The correlation between perceptual tasks and higher-order visual goes to the common variance in the perceptual tasks.

The general visual factor correlated .58 with cognitive performance. But what is the general visual factor? Technically, it is the variance that is shared among the six perceptual tasks. But where this variance comes from is unclear and not distinguishable within the current experimental design. Here are some possibilities: 1. this general visual factor may reflect the overall perceptual ability, including the ability to extract target features from noise in both spatial and temporal contexts (Lu & Dosher, 2008). 2. Alternatively, this general visual factor may reflect attentional control processes (Tsukahara et al., 2020). Individuals who are good at staying on tasks in the presence of distractions may perform well on all visual tasks and on higher-order cognitive tasks. 3. Finally, as is often the case in individual-difference studies with perceptual and cognitive tasks, the general visual factor may reflect motivation, engagement, or test-taking strategies. Of note, these social-psychological factors can not explain the present of stimulus factor. These possibilities are not mutually exclusive, and our design was not built to distinguish among them. Clearly, future designs may be able to disentangle the meanings of this general visual factor.

Appendix A: Hierarchical Psychometric Model

Model specification. Let $i = 1, \ldots, I$ denote individuals, $j \in \{1, 2, 3, 4, 5, 6\}$ denote the six masking tasks, and $k = 1, \ldots, K_{ij}$ denote trials. Let y_{ijk} denote the binary correctness outcome on the kth trial of individual i in masking task j. The data level is given by

$$y_{ijk} \sim \text{Bernoulli}(\Phi(\beta_j\,(s_{ijk} + \theta_{ij}))\,,$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function. The variable s_{ijk} denotes the stimulus strength on trial k for individual i on task j, which was log-transformed and then standardized across all trials and individuals of task j. Standardization places the six masking tasks on a common scale, so that the same priors can be used for all tasks; because it is a linear transformation within each task, it does not affect the correlations among tasks. The parameter θ_{ij} denotes the latent ability of individual i on task j, expressed on the same scale as s_{ijk}, and β_j denotes the rate of the psychometric function for task j.

At the latent level, each individual is characterized by the vector of abilities on the ten tasks, $\boldsymbol{\gamma}_i = (\theta_{i1}, \ldots, \theta_{i6}, c_{i1}, \ldots, c_{i4})'$, where c_{im} is the observed standardized score of individual i on cognitive task m. This vector is generated by the measurement and structural levels described in the main text,

$$\begin{aligned}
\boldsymbol{\gamma}_i &= \boldsymbol{\mu} + \boldsymbol{\Lambda}\,\boldsymbol{\eta}_i + \boldsymbol{\epsilon}_i, \qquad \boldsymbol{\epsilon}_i \sim \mathcal{N}_{10}(\mathbf{0}, \boldsymbol{\Delta})\,, \qquad \boldsymbol{\Delta} = \text{diag}\Big(\delta_1^2, \ldots, \delta_{10}^2\Big)\,, \\
\boldsymbol{\eta}_i &= \mathbf{B}\,\boldsymbol{\eta}_i + \boldsymbol{\zeta}_i, \qquad \boldsymbol{\zeta}_i \sim \mathcal{N}_7(\mathbf{0}, \mathbf{I}_7)\,,
\end{aligned}$$

with $\boldsymbol{\Lambda}$ and $\mathbf{B}$ as given in the main text. The latent vector $\boldsymbol{\eta}_i = (\eta_{i1}, \ldots, \eta_{i7})'$ contains the general visual factor (η_{i1}), the two masking factors (η_{i2}, η_{i3}), the three stimulus factors ($\eta_{i4}, \eta_{i5}, \eta_{i6}$), and the cognitive factor (η_{i7}). The thresholds θ_{ij} are latent and informed by the trial-level data, whereas the cognitive scores enter as observed values. Each cognitive score is therefore modeled through the measurement level as

$c_{im} = \mu_{6+m} + \lambda_{6+m,7}\,\eta_{i7} + \epsilon_{i,6+m}$, so that the variance of each cognitive task that is not shared with the cognitive factor is captured by its unique variance δ^2_{6+m}. Task datasets excluded under the criteria in the Method section contributed no trials to the likelihood; the corresponding thresholds were treated as missing and inferred from the latent structure.

Prior specification. Priors are needed for the task means $\boldsymbol{\mu}$, the slopes β_j, the free factor loadings in $\boldsymbol{\Lambda}$, the unique standard deviations in $\boldsymbol{\Delta}$, and the regression weights in $\mathbf{B}$. The task means and slopes were given weakly informative normal priors, which are wide relative to the standardized scale of s_{ijk} and of the cognitive scores. The free loadings were given half-normal priors, which fix the sign of each factor; all other elements of $\boldsymbol{\Lambda}$ were fixed at zero. The unique standard deviations were given half-t priors, and the six free regression weights in $\mathbf{B}$ were given standard normal priors:

$$
\begin{aligned}
\mu_t &\sim \mathcal{N}(0, 5^2), && t = 1, \ldots, 10,\\
\beta_j &\sim \mathcal{N}(0, 5^2), && j = 1, \ldots, 6,\\
\lambda_{tl} &\sim \mathcal{N}^{+}(0, 1), && \text{for each free loading } \lambda_{tl} \text{ in } \boldsymbol{\Lambda},\\
\delta_t &\sim \text{Half-}t_3(0, 2.5), && t = 1, \ldots, 10,\\
b_l &\sim \mathcal{N}(0, 1), && l = 1, \ldots, 6.
\end{aligned}
$$

Implied covariance. The covariance matrix of $\boldsymbol{\gamma}_i$ is not a free parameter but is implied by the loadings, the regression weights, and the unique variances,

$$\boldsymbol{\Sigma} = \boldsymbol{\Lambda}\,(\mathbf{I}_7 - \mathbf{B})^{-1}\,(\mathbf{I}_7 - \mathbf{B})^{-\top}\,\boldsymbol{\Lambda}^{\top} + \boldsymbol{\Delta}.$$

We computed $\boldsymbol{\Sigma}$ and the corresponding correlation matrix for each posterior draw; the model-estimated correlations reported in the main text are their posterior means.

Bayes factors. The Bayes factor for the hypothesis that the lth non-cognitive factor is unrelated to cognitive performance compares a model in which b_l is free to a nested

model in which $b_l = 0$, with all other parameters unchanged. Because the nested model is obtained by fixing a single parameter, the Bayes factor equals the Savage–Dickey density ratio (Dickey & Lientz, 1970),

$$\mathrm{BF}_{10} = \frac{p(b_\ell = 0)}{p(b_\ell = 0 \mid \text{data})},$$

where the numerator is the prior density at zero, $1/\sqrt{2\pi}$, and the denominator was estimated from MCMC outputs using the CMDE method of (Morey et al., 2011)

Model fitting. The model was implemented in JAGS (Plummer, 2003) and run from R with the R2jags package. We ran 4 chains, each with 50,000 burn-in steps followed by 400,000 sampling steps. Chains were thinned by a factor of 100, yielding 16,000 posterior draws. Convergence was assessed with the Gelman–Rubin $\hat{R}$ statistic (Gelman & Rubin, 1992), which was below 1.01 for all parameters.

Appendix B: Derivation of the Variance Decomposition

This appendix derives the two quantities reported in the main text: the proportion of variance of each masking task due to each source, and the correlation of each non-cognitive factor with the cognitive factor. Here and below, the index $l = 1, \ldots, 6$ runs over the six non-cognitive factors. Both quantities follow from the model specified in Appendix A. As a reminder, the structural level is given by

$$\boldsymbol{\eta}_i = \mathbf{B}\,\boldsymbol{\eta}_i + \boldsymbol{\zeta}_i, \qquad \boldsymbol{\zeta}_i \sim \mathcal{N}_7(\mathbf{0}, \mathbf{I}_7)\,.$$

Because only the last row of $\mathbf{B}$ is nonzero, the cognitive factor is the only factor that receives paths from other factors. Each non-cognitive factor is therefore equal to its own residual, $\eta_{il} = \zeta_{il}$, whereas the cognitive factor is a weighted sum of the non-cognitive factors plus its own residual, $\eta_{i7} = \sum_{l=1}^{6} b_l\,\eta_{il} + \zeta_{i7}$. The variances and covariances are

therefore

$$\mathrm{Var}(\eta_{i\ell}) = 1, \qquad \mathrm{Cov}(\eta_{i\ell}, \eta_{i\ell'}) = 0 \;\; (\ell \neq \ell'), \qquad \mathrm{Cov}(\eta_{i\ell}, \eta_{i7}) = b_l,$$
$$\mathrm{Var}(\eta_{i7}) = \sum_{\ell=1}^{6} b_\ell^2 \; \mathrm{Var}(\eta_{i\ell}) + \mathrm{Var}(\zeta_{i7}) = \sum_{\ell=1}^{6} b_\ell^2 + 1.$$

Variance decomposition of the masking tasks. The threshold on masking task j is given by the jth row of the measurement level,

$$\theta_{ij} = \mu_j + \sum_{l=1}^{6} \lambda_{jl}\, \eta_{il} + \epsilon_{ij},$$

where only three of the six loadings $\lambda_{j1}, \ldots, \lambda_{j6}$ are nonzero: the loading on the general visual factor, λ_{j1}, the loading on the factor for the task's masking procedure (one of λ_{j2} and λ_{j3}), and the loading on the factor for the task's stimulus (one of λ_{j4}, λ_{j5}, and λ_{j6}). Because the unique term ϵ_{ij} is independent of the factors, the variance of θ_{ij} is

$$\mathrm{Var}(\theta_{ij}) = \sum_{l=1}^{6} \sum_{l'=1}^{6} \lambda_{jl}\lambda_{jl'}\, \mathrm{Cov}(\eta_{il}, \eta_{il'}) + \delta_j^2.$$

Given that the covariances among the non-cognitive factors are zero and their variances are fixed to one, all cross-product terms vanish, and the above equation simplifies to

$$\mathrm{Var}(\theta_{ij}) = \sum_{l=1}^{6} \lambda_{jl}^2 + \delta_j^2 = \underbrace{\lambda_{j1}^2}_{\text{general}} + \underbrace{\lambda_{j2}^2 + \lambda_{j3}^2}_{\text{masking}} + \underbrace{\lambda_{j4}^2 + \lambda_{j5}^2 + \lambda_{j6}^2}_{\text{stimulus}} + \underbrace{\delta_j^2}_{\text{unique}}.$$

For example, for the first task, temporal masking with the Pi stimulus, $\mathrm{Var}(\theta_{i1}) = \lambda_{11}^2 + \lambda_{12}^2 + \lambda_{14}^2 + \delta_1^2$. Dividing each component by the total variance gives the proportion of variance due to each source.

Contribution of each non-cognitive factor to cognitive performance. Given $\mathrm{Var}(\eta_{i7}) = \sum_{\ell=1}^{6} b_\ell^2 + 1$ and $\mathrm{Cov}(\eta_{i\ell}, \eta_{i7}) = b_\ell$, the correlation between the lth non-cognitive

factor and the cognitive factor is

$$r_\ell = \frac{\mathrm{Cov}(\eta_{i\ell}, \eta_{i7})}{\sqrt{\mathrm{Var}(\eta_{i\ell})\,\mathrm{Var}(\eta_{i7})}} = \frac{b_\ell}{\sqrt{\sum_{\ell'=1}^{6} b_{\ell'}^2 + 1}},$$

which was computed for each posterior draw; Table 2 reports its posterior mean and 95% credible interval.

Declarations

Funding: JNR was supported by ONR N00014-23-1-2792.

Conflict of Interest: None.

Ethical Approval: Something

Consent to Participate: Something

Consent for Publication: Something

Data Availability: Something

Code Availability: All data and code required to reproduce the analyses are available at .

Author Contributions: Something

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process:

During the preparation of this work the authors used Claude and ChatGPT in order to check the grammar and spelling, as well as for verifying some mathematical calculations. After using this tool/service, the authors reviewed and edited the content as needed and took full responsibility for the content of the published article.

Reference

Bollen, K. A. (1989). *Structural equations with latent variables*. Wiley.

Conway, A. R., Kane, M. J., Bunting, M. F., Hambrick, D. Z., Wilhelm, O., & Engle, R. W. (2005). Working memory span tasks: A methodological review and user's guide. *Psychonomic Bulletin & Review*, *12*, 769–786.

Deary, I. J., Bell, P. J., Bell, A. J., Campbell, M. L., & Fazal, N. D. (2004). Sensory Discrimination and Intelligence: Testing Spearman's Other Hypothesis. *Am. J. Psychol.*, *117*(1), 1. doi:10.2307/1423593

Deary, I. J., & Stough, C. (1996). Intelligence and inspection time: Achievements, prospects, and problems. *American Psychologist*, *51*(6), 599.

Dickey, J. M., & Lientz, B. P. (1970). The Weighted Likelihood Ratio, Sharp Hypotheses about Chances, the Order of a Markov Chain. *The Annals of Mathematical Statistics*, *41*(1), 214–226. Retrieved from http://www.jstor.org/stable/2239734

Galton, F. (1883). *Inquiries into human faculty and its development* (pp. xii, 387). New York, NY, US: MacMillan Co. doi:10.1037/14178-000

Gelman, A., & Rubin, D. B. (1992). Inference from iterative simulation using multiple sequences (with discussion). *Statistical Science*, *7*, 457–511.

Grudnik, J. L., & Kranzler, J. H. (2001). Meta-analysis of the relationship between intelligence and inspection time. *Intelligence*, *29*(6), 523–535.

Jastrzębski, J., Kroczek, B., & Chuderski, A. (2021). Galton and Spearman revisited: Can single general discrimination ability drive performance on diverse sensorimotor tasks and explain intelligence? *Journal of Experimental Psychology: General*, *150*(7), 1279–1302. doi:10.1037/xge0001005

Jensen, A. R. (2006). *Clocking the mind: Mental chronometry and individual differences*. Elsevier.

Kranzler, J. H., & Jensen, A. R. (1989). Inspection time and intelligence: A meta-analysis. *Intelligence*, *13*(4), 329–347. doi:10.1016/S0160-2896(89)80006-6

Levitt, H. (1971). Transformed Up-Down Methods in Psychoacoustics. *The Journal of the Acoustical Society of America*, *49*, 467–477. doi:10.1121/1.1912375

Lu, Z.-L., & Dosher, B. A. (2008). Characterizing observers using external noise and observer models: Assessing internal representations with external noise. *Psychological Review*, *115*(1), 44–82. doi:10.1037/0033-295X.115.1.44

Mikellidou, K., Lambrou, N., Georgiou, E., & Avraamides, M. (2023). Visual orientation discrimination skills are tightly linked with specific aspects of human intelligence. *PLOS ONE*, *18*(10), e0289590. doi:10.1371/journal.pone.0289590

Morey, R. D., Rouder, J. N., Pratte, M. S., & Speckman, P. L. (2011). Using MCMC chain outputs to efficiently estimate Bayes factors. *Journal of Mathematical Psychology*, *55*, 368–378. Retrieved from http://dx.doi.org/10.1016/j.jmp.2011.06.004

Mosing, M. A., Pedersen, N. L., Madison, G., & Ullén, F. (2014). Genetic Pleiotropy Explains Associations between Musical Auditory Discrimination and Intelligence. *PLOS ONE*, *9*(11), e113874. doi:10.1371/journal.pone.0113874

Nettelbeck, T., & Lally, M. (1976). INSPECTION TIME AND MEASURED INTELLIGENCE. *Br. J. Psychol.*, *67*(1), 17–22. doi:10.1111/j.2044-8295.1976.tb01493.x

Pahor, A., Seitz, A. R., & Jaeggi, S. M. (2022). Near transfer to an unrelated N-back task mediates the effect of N-back working memory training on matrix reasoning. *Nat. Hum. Behav.*, *6*(9), 1243–1256. doi:10.1038/s41562-022-01384-w

Pahor, A., Stavropoulos, T., Jaeggi, S. M., & Seitz, A. R. (2019). Validation of a matrix reasoning task for mobile devices. *Behav. Res. Methods*, *51*(5), 2256–2267. doi:10.3758/s13428-018-1152-2

Parish, D. H., & Sperling, G. (1991). Object spatial frequencies, retinal spatial frequencies, noise, and the efficiency of letter discrimination. *Vision Research*, *31*(7), 1399–1415. doi:10.1016/0042-6989(91)90060-I

Peirce, J., Gray, J. R., Simpson, S., MacAskill, M., Höchenberger, R., Sogo, H., ...

Lindeløv, J. K. (2019). PsychoPy2: Experiments in behavior made easy. *Behavior Research Methods*, *51*(1), 195–203. doi:10.3758/s13428-018-01193-y

Plummer, M. (2003). JAGS: A Program for Analysis of Bayesian Graphical Models Using Gibbs Sampling. In *Proceedings of the 3rd International Workshop on Distributed Statistical Computing*.

Robison, M. K., Campbell, S., Garner, L. D., Sibley, C., & Coyne, J. (2026). A comprehensive psychometrics of cognitive ability measures: Reliability, practice effects, and the stability of latent factor structures across retesting. *Behavior Research Methods*, *58*(2), 56. doi:10.3758/s13428-025-02897-8

Rouder, J. N., & Haaf, J. M. (2019). A psychometrics of individual differences in experimental tasks. *Psychonomic Bulletin and Review*, *26*(2), 452–467. Retrieved from https://doi.org/10.3758/s13423-018-1558-y

Rouder, J. N., & Mehrvarz, M. (2026). Random-effects psychophysics for studying individual differences in perception and cognition. *Psychon. Bull. Rev.*, *33*(5), 144. doi:10.3758/s13423-026-02890-y

Salthouse, T. A. (1996). The processing speed theory of adult age differences in cognition. *Psychological Review*, *103*, 403–428.

Sokal, R. R., & Michener, C. D. (1958). A statistical method for evaluating systematic relationships. *University of Kansas Science Bulletin*, *38*, 1409–1438.

Spearman, C. (1904). 'General intelligence,' objectively determined and measured. *The American Journal of Psychology*, *15*(2), 201–293. doi:10.2307/1412107

Tsukahara, J. S., Harrison, T. L., Draheim, C., Martin, J. D., & Engle, R. W. (2020). Attention control: The missing link between sensory discrimination and intelligence. *Attention, Perception, & Psychophysics*, *82*(7), 3445–3478. doi:10.3758/s13414-020-02044-9

Unsworth, N., Heitz, R. P., Schrock, J. C., & Engle, R. W. (2005). An automated version of the operation span task. *Behavior Research Methods*, *37*(3), 498–505.

doi:10.3758/BF03192720

Vernon, P. A. (1983). Speed of information processing and general intelligence. *Intelligence*, *7*(1), 53–70. doi:10.1016/0160-2896(83)90006-5

Vickers, D., Nettelbeck, T., & Willson, R. J. (1972). Perceptual indices of performance: The measurement of "inspection time" and "noise" in the visual system. *Perception*, *1*(3), 263–295. doi:10.1068/p010263

Table 1
Proportion of variance of each masking task due to each source.

Task	Mask type	Stimulus type	General	Masking	Stimulus	Unique
1	Temporal	Pi	.30 [.06, .64]	.04 [.00, .19]	.45 [.07, .82]	.21 [.00, .52]
2	Temporal	Letter	.17 [.01, .46]	.25 [.01, .65]	.39 [.05, .79]	.19 [.00, .58]
3	Temporal	Gabor	.28 [.04, .60]	.22 [.00, .66]	.20 [.00, .64]	.29 [.00, .67]
4	Spatial	Pi	.33 [.05, .70]	.15 [.00, .48]	.34 [.05, .70]	.18 [.00, .46]
5	Spatial	Letter	.36 [.07, .72]	.18 [.00, .56]	.28 [.02, .64]	.18 [.00, .51]
6	Spatial	Gabor	.48 [.12, .87]	.09 [.00, .42]	.20 [.00, .64]	.23 [.00, .57]
Average			**.32 [.19, .45]**	**.16 [.07, .26]**	**.31 [.20, .43]**	**.21 [.12, .32]**

Note. Values are posterior means with 95% credible intervals.

Table 2
Correlation of each non-cognitive factor with cognitive performance.

Factor	r	BF_{10}
General visual factor	.56 [.15, .83]	**14.0**
Temporal masking	-.03 [-.43, .40]	0.44
Spatial masking	.20 [-.27, .61]	0.69
Pi stimulus	.17 [-.19, .50]	0.58
Letter stimulus	-.16 [-.50, .23]	0.55
Gabor stimulus	.27 [-.25, .67]	0.97

Note. r is the posterior mean of the correlation r_l between each non-cognitive factor and the cognitive factor, with 95% credible interval. Bayes factors are Savage–Dickey density ratios with prior $b_l \sim \mathcal{N}(0, 1)$. BF_{10} values above 3 provide positive evidence for a relation, and values above 20 provide strong evidence.

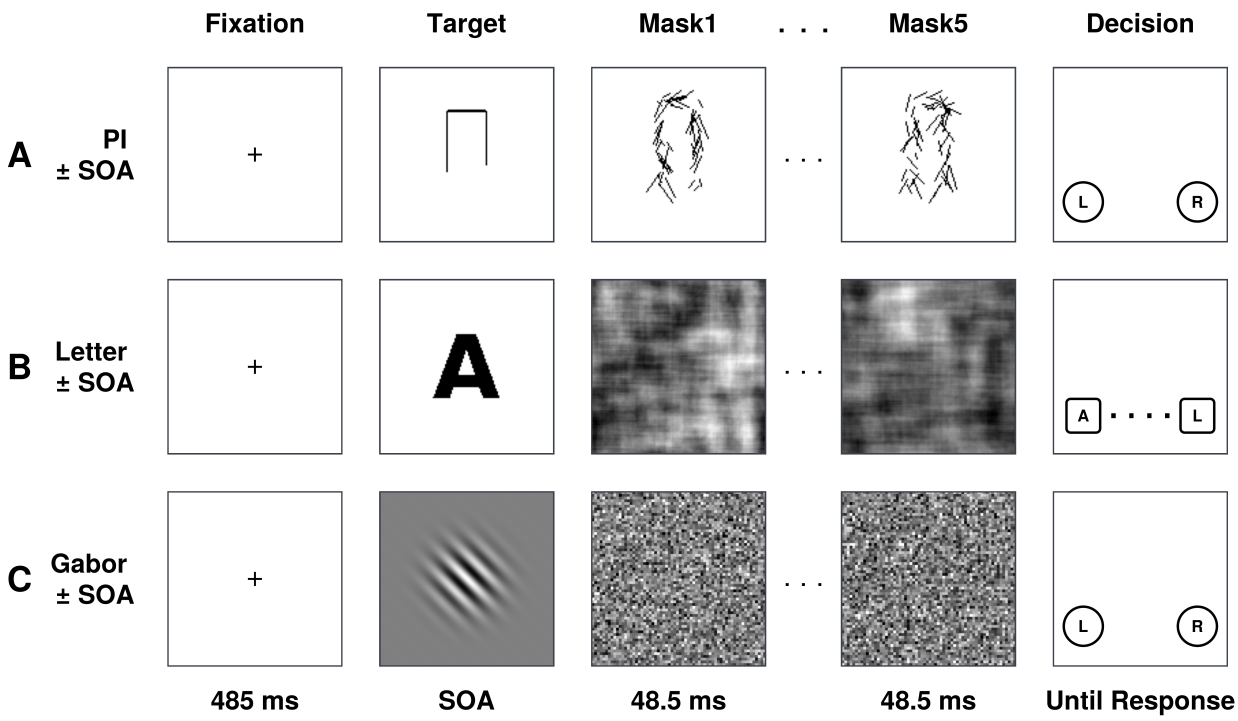


Figure 1. Trial sequence of the three temporal masking tasks. Rows show the Pi, Letter, and Gabor tasks. Each trial began with a fixation cross (485 ms), followed by the target. After a variable stimulus onset asynchrony (SOA), the target was replaced by a sequence of five masks (48.5 ms each): line segments for the Pi task, frequency-matched letter noise for the Letter task, and white noise for the Gabor task. Individuals then judged which leg of the Pi was longer (left or right), which of nine letters (A-L) was presented, or whether the Gabor was tilted to the left or right. The response screen remained until response. Task difficulty was adjusted by varying the SOA.

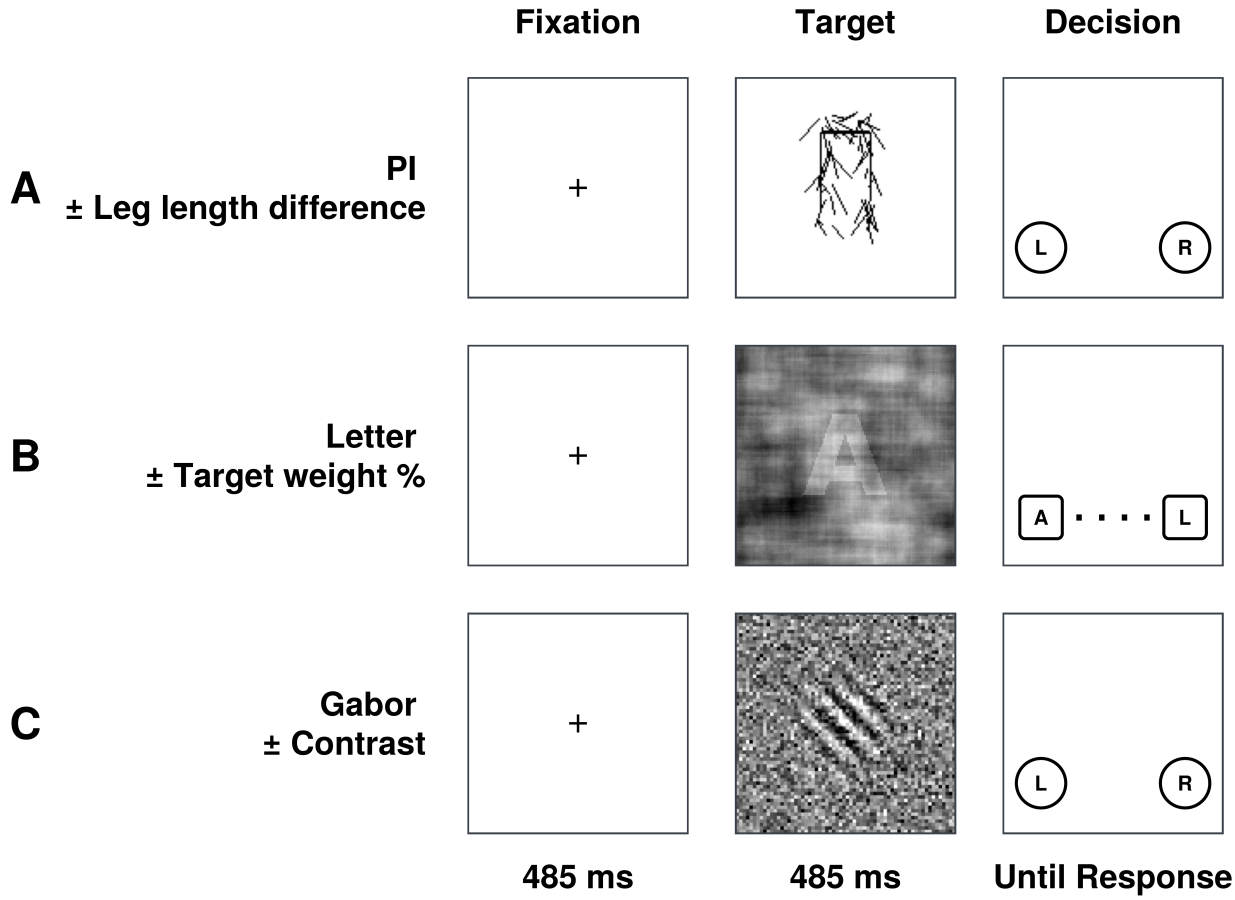


Figure 2. Trial sequence of the three spatial masking tasks. Rows show the Pi, Letter, and Gabor tasks. Each trial began with a fixation cross (485 ms), followed by the target embedded in the same mask as in the corresponding temporal masking task (485 ms): line segments for the Pi task, frequency-matched letter noise for the Letter task, and white noise for the Gabor task. Individuals made the same judgments as in the temporal masking tasks, and the response screen remained until response. Task difficulty was adjusted by varying the leg-length difference of the Pi, the target weight in the letter-noise blend, and the contrast of the Gabor, respectively.

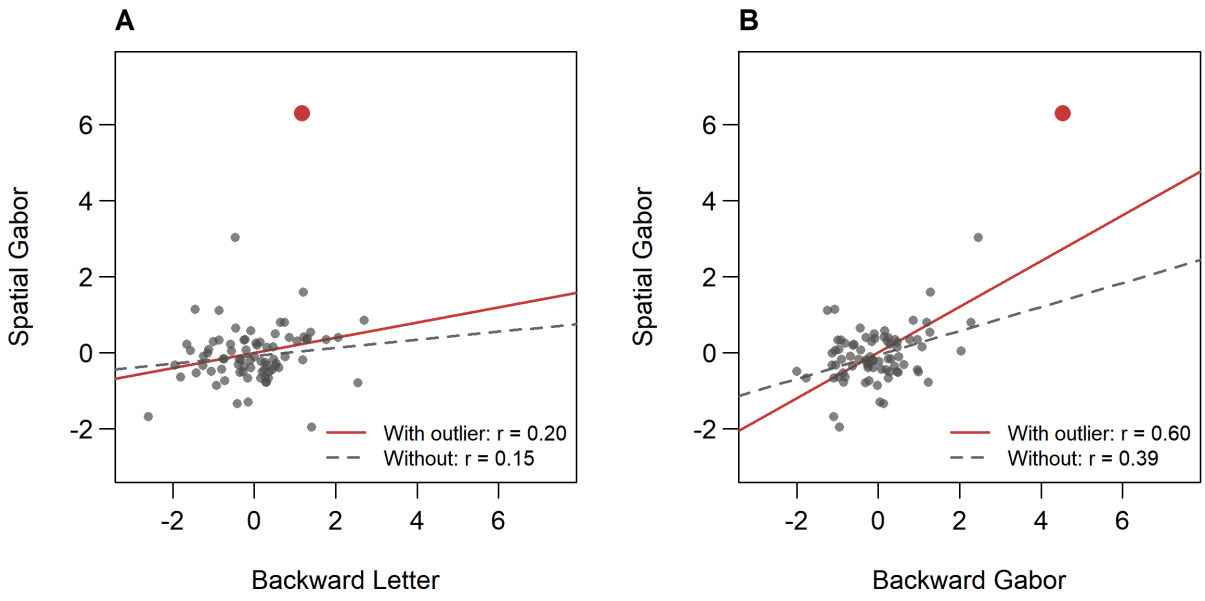


Figure 3. Something

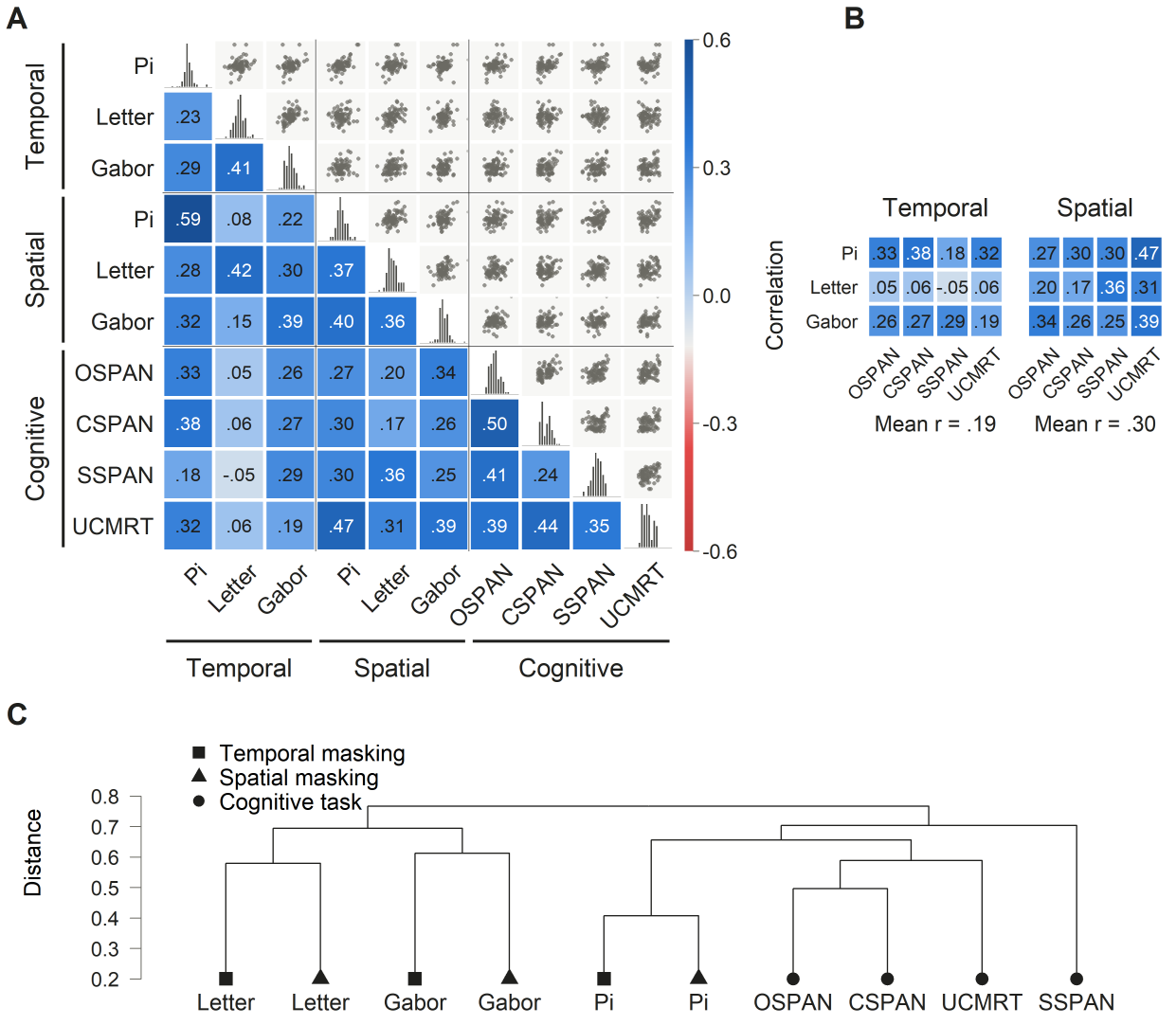


Figure 4. Something

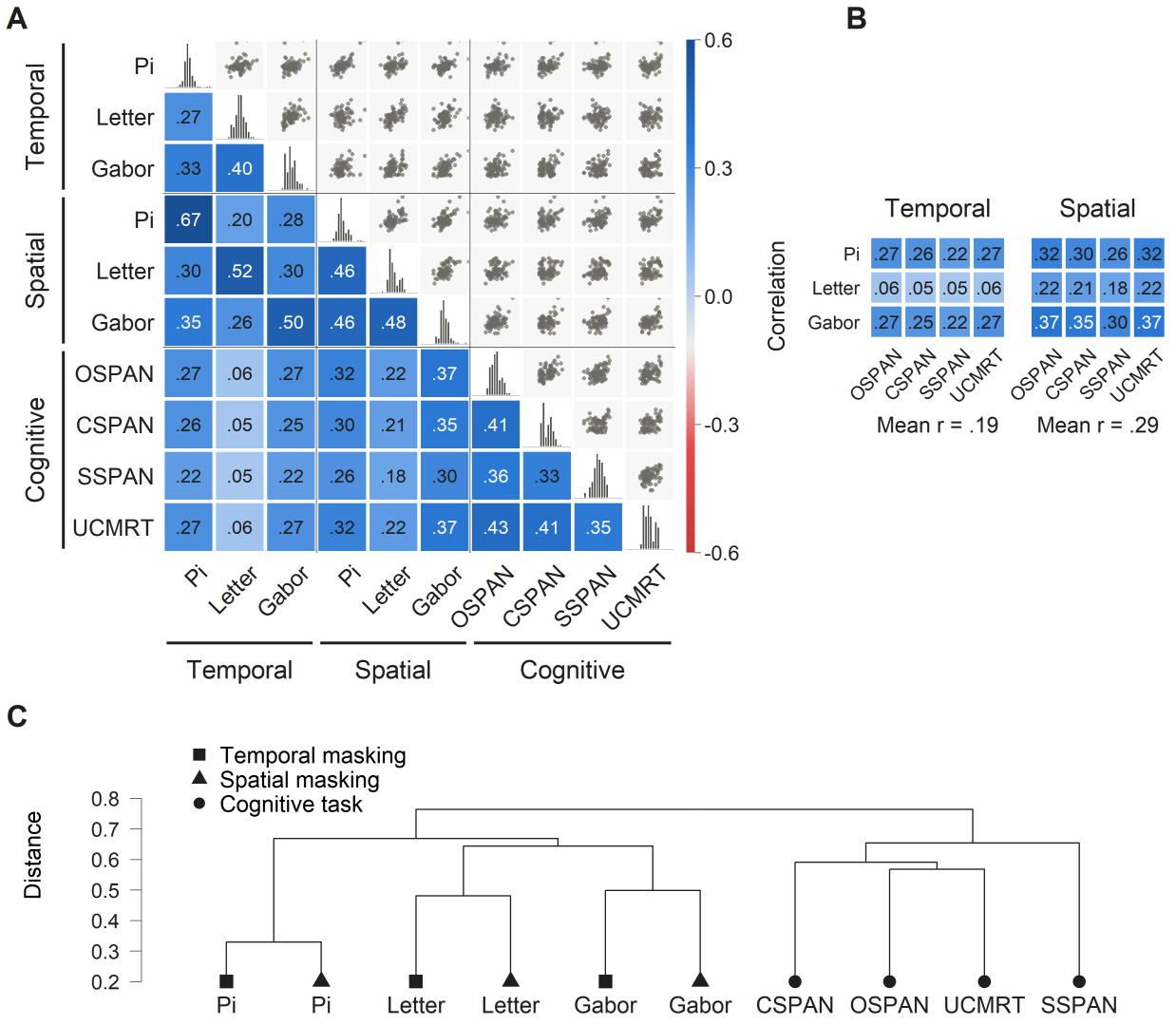


Figure 5. Something